

Project #2 Autonomous Underwater Vehicle (AUV)

Sensor-Fusion-Based Perception, Localization, Navigation and Autonomous Mission Execution

Graduation Project Description - Continuation of the Ain Shams University ASmarine AUV Programme

1. Project Description

This project continues the Autonomous Underwater Vehicle (AUV) development carried out during previous graduation projects at Ain Shams University and demonstrated in the referenced ASmarine videos. The earlier work established the mechanical platform, pressure housing, propulsion arrangement, electrical modules, underwater perception experiments, SLAM development, and nonlinear control studies. The new project will transform these subsystems into one integrated, mission-ready vehicle that can operate autonomously in GPS-denied underwater environments.

The target system shall perform underwater exploration, structural inspection, target search, visual and acoustic mapping, waypoint following, station keeping, obstacle avoidance, and safe return or recovery. During normal autonomous missions, the AUV should not depend on a tether; a surface Ethernet/USB connection may be retained for laboratory development, software debugging, charging, mission upload, and supervised testing.

CAD MODEL / SYSTEM DESIGN



REAL ASMARINE PROTOTYPE



Figure 1. Previous ASmarine AUV mechanical design and the real prototype developed at Ain Shams University.

PROJECT AIM

Develop and validate a fully autonomous AUV that combines multi-sensor perception, reliable underwater localization, intelligent mission planning, robust six-degree-of-freedom control, and safe propulsion/power management.

CORE OBJECTIVES

- Integrate camera, DVL, sonar, IMU, depth and heading sensors.
- Fuse sensor data for drift-reduced 6-DOF state estimation.
- Implement SLAM, mapping, waypoint navigation and obstacle avoidance.
- Implement station keeping, trajectory tracking and return-to-home.
- Validate performance through simulation, pool tests and mission trials.

Architecture summary. Raw measurements from the vision, acoustic, inertial, depth and velocity sensors are time-synchronised and fused using an EKF or equivalent estimator. The estimated pose and map feed the navigation and path-planning modules. Guidance and control generate desired forces and moments, which are converted by the thruster-allocation algorithm into individual motor commands. Mission data, images, maps, alarms and performance indicators are logged onboard and displayed at the surface station during supervised tests.

2. Proposed System Layout, Sensors and Autonomy Architecture

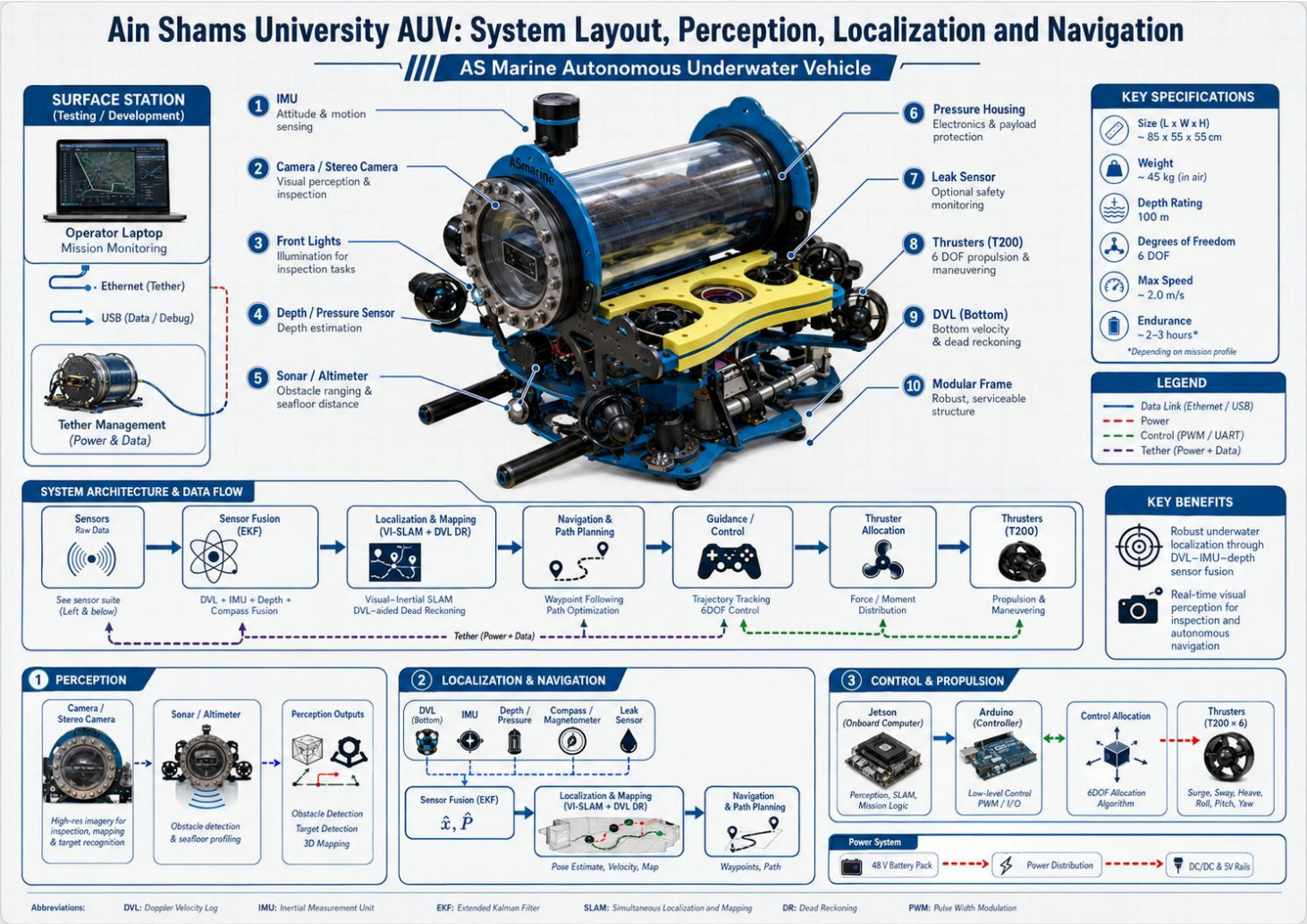


Figure 2. Proposed AUV system layout and data flow for perception, localization, navigation, control and propulsion.

3. Main Components to Be Included

Subsystem	Required Components	Function in the AUV
Mechanical and pressure system	Existing modular frame; sealed electronics pressure housing; buoyancy and ballast elements; camera/light mounts; DVL and sonar mounting; serviceable connectors and cable routing.	Protect payloads, maintain trim and stability, and allow rapid maintenance or sensor replacement.
Perception sensors	Stereo/depth camera with underwater calibration and refraction correction; front lights; forward or 360-degree imaging sonar; altimeter/short-range sonar.	Visual inspection, target detection, obstacle awareness, range estimation and local mapping.
Localization sensors	DVL; IMU; magnetometer/compass; pressure/depth sensor; optional surface GNSS for initialisation and recovery.	Estimate position, velocity, attitude, heading and depth in GPS-denied operation.
Onboard computing	NVIDIA Jetson or equivalent embedded computer; real-time microcontroller such as Arduino/STM32; onboard storage; ROS-based middleware.	Run perception, fusion, SLAM, mission logic, control, diagnostics and data logging.
Propulsion and actuation	Six T200-class thrusters or the validated existing thruster configuration; ESCs; low-level PWM/serial interface; thrust-characterisation model.	Provide surge, sway, heave, roll, pitch and yaw control and distribute commanded forces/moments.
Power, safety and communication	Battery pack; BMS; fuses and isolated DC/DC rails; leak sensors; emergency stop/kill logic; development tether and surface operator laptop.	Provide protected power, fault monitoring, safe shutdown, mission upload, telemetry and supervised override.

4. Development Scope and Work Packages

WP1 - Existing Platform Review and Mechanical Upgrade	Inspect the previous AUV, verify frame integrity and sealing, rebalance buoyancy/ballast, improve mounts for the DVL, sonar and camera, and revise cable routing for maintainability.
WP2 - Electrical, Power and Safety Integration	Validate the battery/BMS and distribution board; integrate protected voltage rails, leak detection, emergency stop, watchdogs, fault isolation, status indicators and safe start-up/shutdown logic.
WP3 - Sensor Calibration and Synchronisation	Calibrate camera intrinsics/extrinsics underwater; compensate refraction; calibrate IMU, magnetometer, DVL, depth and sonar; create common timestamps and sensor-health diagnostics.
WP4 - Perception, Localization and Mapping	Develop visual/acoustic perception, object/target detection, obstacle ranging, EKF-based sensor fusion, DVL-aided dead reckoning, visual-inertial SLAM and map generation.
WP5 - Autonomous Navigation and Control	Implement mission state machine, waypoint following, depth/heading control, station keeping, path planning, obstacle avoidance, thruster allocation and robust/nonlinear trajectory control.
WP6 - Operator Interface, Data Logging and Evaluation	Develop a desktop interface for mission planning and telemetry; store synchronized sensor data, video, maps and alarms; perform simulation, pool trials and final underwater mission demonstrations.



Figure 3. Representative autonomous mission sequence from planning and self-check to recovery and reporting.

Previous ASmarine project videos: [Video 1](#) | [Video 2](#) | [Video 3](#)

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